

Lecture 14 DAA Mar 4

Given a knapsack of capacity B
and n items $1, 2, \dots, n$
with wts $w_1, w_2, \dots, w_n$
and profits $p_1, p_2, \dots, p_n$

$$\text{maximize } \sum p_i \cdot x_i$$

$$\text{s.t. } \sum w_i \cdot x_i \leq B$$

$$x_i \in \{0, 1\} \quad \text{! if item is chosen}$$

Example

Profits
wts

10	10	12	18
2	4	6	9

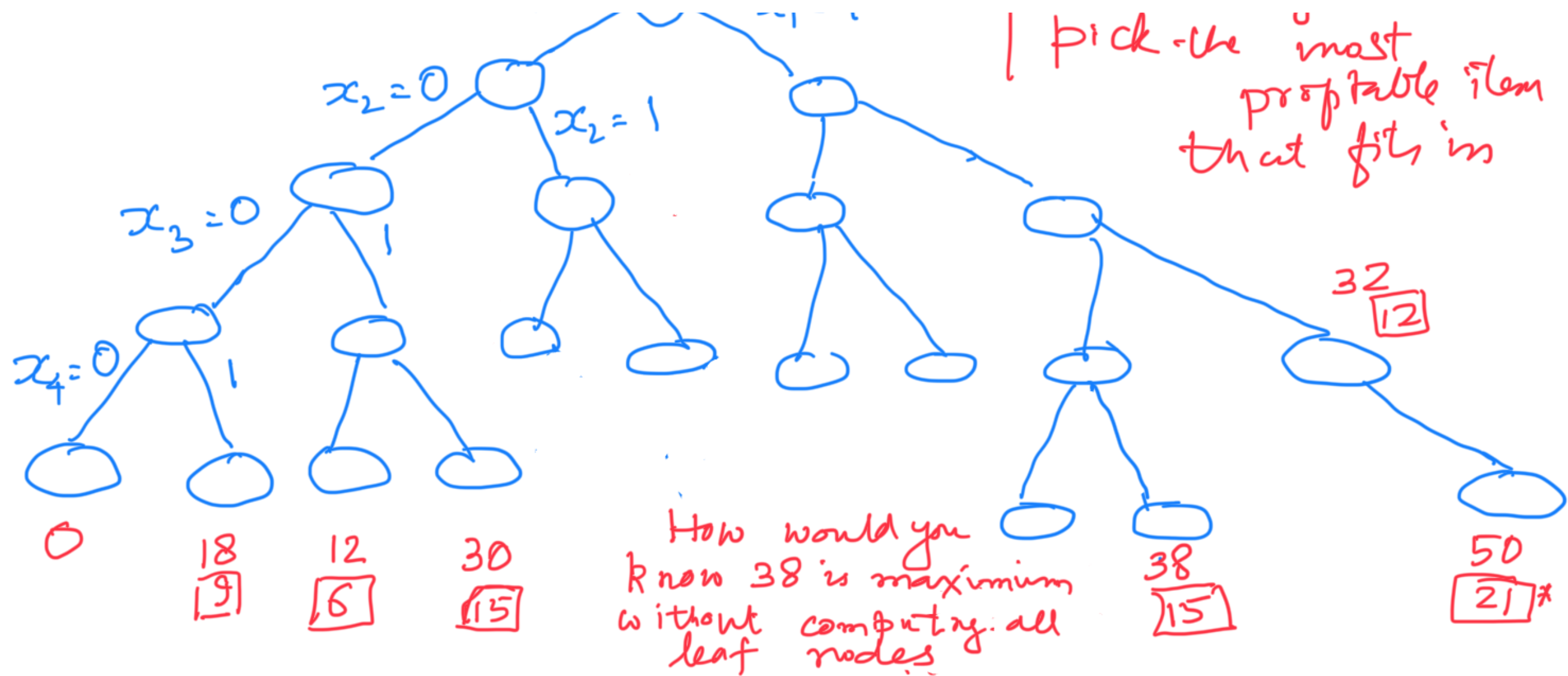
5 2.5 2 2 $B = 15$

Greedy by ratio
may not work

rather prof/wt

$x_i = 0$ or $x_i = 1$

Greedy would



Greedy gives a profit $30 = (18 + 12)$

whereas there is a soln 38 $\begin{matrix} \{3, 4\} \\ \{1, 2, 4\} \end{matrix}$

Clearly there is an instance of profit, weights where greedy will not provide the best soln

The algorithm that we are seeking must

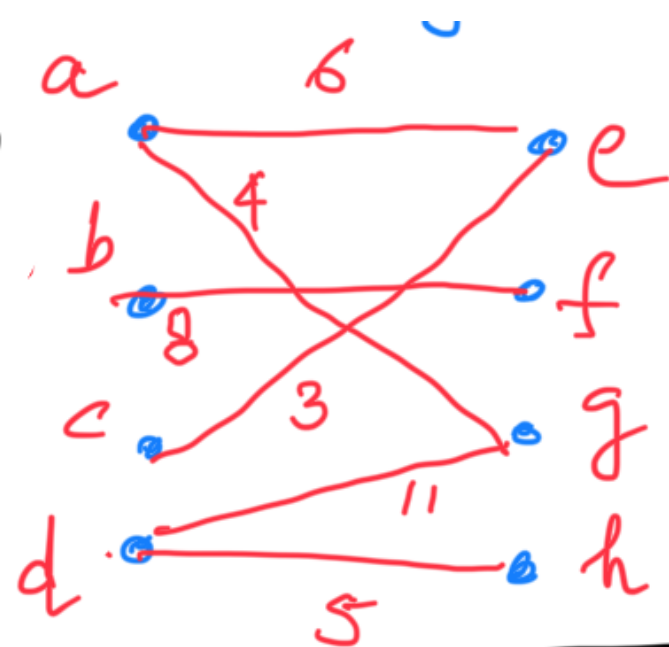
(1) $\forall$ choices wts, $\forall$ profits and B
it must return the best soln

(2) Must be efficient

Maxm Wt ^(N x N) Matching in bipartite graphs

Matching
 $\{(a,e) (b,f) (d,g)\}$
 $\begin{matrix} 6 & 8 & 11 \\ = 25 \end{matrix}$

$\{(a,g), (b,f) (e,e), (d,h)\} = 20$



Objective is to select edges whose ^{sum of} weights will be maximum

subject to :

the set of valid edges is called Matching

No two edges in among the chosen subset have vertices in common

For example (a,g) and (a,e) cannot be chosen in the "matching"

If every vertex in the graph belongs to a matched edge, then it is perfect Matching

A possible soln to matching in the above example is $\{(a,e) (b,f)\}$

Observation : 1. Any subset of a valid matching is also valid

2. Any subset of a feasible soln for Knapsack is also feasible

3. (MST problem) Any subset of a forest is also a forest, i.e. no cycles

The above property is called a Subset System

Any soln of a feasible soln is feasible

Maximal feasible subset

A subset is maximal if no further element can be added without violating feasibility.

For some problem (like MST), the
size of the maximal subsets are identical
(but not so for Knapsack, MWM)

Greedy framework

Given a set $S = \{e_1, e_2, \dots, e_m\}$
with weights $w_1, w_2, \dots, w_m$

(assume $w_1 \geq w_2 \geq \dots \geq w_m$

↳ else we can sort in decreasing
order and rename)

M : family of subsets of S , i.e. $M \subseteq 2^S$
that are feasible / independent

Goal: Find the maximum wt subset of M

Greedy Algorithm (S, M^*)

$T = \emptyset$ $\{ \text{initialization} \}$ $\emptyset \in M$

For $i = 1$ to n do

[If $T \cup \{e_i\} \in M$
then $T = T \cup \{e_i\}$
(else discard)

M^* is not explicitly given
so an appropriate test is required

Output T as the optimum subset

When is T guaranteed to be the correct solution

Necessary and sufficient (and equivalent)

1. Exchange property is satisfied

Let S_k, S_{k+1} be both feasible subsets with k and $k+1$ elements

Then $\exists e \in S_{k+1} - S_k$ s.t. $S_k \cup \{e\}$ is feasible

2. Rank property All maximal feasible subsets have the same number of elements

To verify if the greedy algorithm works correctly (for a subset system)

• Either prove property 1 or property 2

Else to show that greedy does not work

- find a counter-example to property 1 or property 2

Intuition : The exchange property ensures that one does not get stuck with the choices made so far and can be extended to a better soln if there exists one

Note that it works for the unweighted version (all wts are 1) and somehow works equally well for the weighted version